

```
class HebbianAttention(nn.Module):
    def __init__(self, dim):
        super().__init__()
        self.W_q =
nn.Parameter(torch.randn(dim, dim) * 0.1)
        self.W_k =
nn.Parameter(torch.randn(dim, dim) * 0.1)

    def forward(self, x):
        Q = x @ self.W_q # Hebbian query
        K = x @ self.W_k # Hebbian key
        # Локальное обновление весов
        delta_Wq = lr * torch.mean(x.T @ Q,
dim=1, keepdim=True)
        self.W_q.data += delta_Wq
        return F.softmax(Q @ K.T / sqrt(d_k),
dim=-1)
```